

Data Extraction Guide — GeoBoundaries, Climate, and Soil (HWSD2)

This README explains **what to run, in which order, what folders to create, and what outputs to expect** for extracting **GeoBoundaries**, **Climate**, and **Soil** features for the Algeria–Tunisia Area of Interest (AOI).

Execution order:

1. `Code/Extracting Data/Boundaries.ipynb`
2. `Code/Extracting Data/Climate.ipynb`
3. `Code/Extracting Data/SoilNewDataset.ipynb`

1) Prerequisites

- **Python** 3.10+ (Anaconda/Miniconda recommended)
- Recommended packages:
 - `geopandas`, `shapely`, `pyproj`
 - `rasterio`, `rioxarray`, `xarray`
 - `numpy`, `pandas`
 - `matplotlib`
 - `pyodbc` (for reading `HWSD2.mdb` on Windows with Access ODBC Driver)
- **GDAL** runtime available (installed automatically with `rasterio` on most conda envs).
- **Data available locally** (example layout below; adjust paths in notebooks if yours differ):

```
Datasets/  
├─ GeoBoundaries/                ← raw boundaries or country shapefiles  
(DZA, TUN)  
├─ ClimateDataset/              ← e.g., WorldClim or similar rasters  
(tmin/tmax/prec)  
└─ SoilDataset/  
    ├─ HWSD2_RASTER/  
    │   ├─ HWSD2.bil  
    │   ├─ HWSD2.hdr  
    │   └─ HWSD2.prj  
    └─ HWSD2_DB/  
        └─ HWSD2.mdb
```

2) Create Output Folders

Create these three folders **before** running the notebooks (Windows PowerShell example):

```
mkdir ExtractedDatasets\GeoBoundaries  
mkdir ExtractedDatasets\ClimateFeatures  
mkdir ExtractedDatasets\SoilFeatures
```

If they already exist, ensure they are empty or that you are okay with overwriting/adding new files.

3) Notebook 1 — **Boundaries.ipynb**

Purpose

- Build a clean **Area of Interest (AOI)** for **Algeria + Tunisia**.
- (Optional) Generate a **regular analysis grid** (e.g., 0.1°) covering the AOI.
- Reproject to a suitable CRS when required.

Expected Inputs

- Country boundaries (Algeria, Tunisia) from your **Datasets/GeoBoundaries** (GeoJSON, SHP, or GPKG).

Key Parameters (typical)

- **GRID_RES_DEG** = **0.1** (≈11 km) or **0.0833** (≈5 arc-min)
- Target CRS (if needed): **EPSG:4326** for lat/lon, or a projected CRS for area-based analyses.

Outputs (written to **ExtractedDatasets/GeoBoundaries/AOI_DZA_TUN/**)

- **aoi_dza_tun.gpkg** — merged polygon of Algeria + Tunisia (primary AOI)
- **aoi_dza_tun.geojson** — same AOI in GeoJSON (handy for web/GIS tools)

Verification tip: Load **aoi_dza_tun.gpkg** in QGIS and ensure the footprint matches both countries precisely (no multipolygon gaps).

4) Notebook 2 — **Climate.ipynb**

Purpose

- Clip **climate rasters** to the AOI (and optionally to the grid).
- Extract **per-cell climate features** such as:
 - **tmin** (minimum temperature)
 - **tmax** (maximum temperature)
 - **prec** (precipitation)
- Optionally compute monthly or annual aggregates and **zonal stats** per grid cell.

Expected Inputs

- Climate rasters in **Datasets/ClimateDataset** (e.g., WorldClim or similar).
- AOI and/or grid produced by **Notebook 1**.

Key Parameters (typical)

- Variables: **tmin**, **tmax**, **prec**
- Temporal aggregation: monthly stacks → annual mean/sum (optional)

- Resampling: `nearest/bilinear` when aligning to grid resolution

Outputs (written to `ExtractedDatasets/ClimateFeatures/`)

- `grid_0p1.gpkg` — analysis grid (0.1°) used in processing
- `tmin_2020_2024.csv` — extracted minimum temperature features
- `tmax_2020_2024.csv` — extracted maximum temperature features
- `prec_2020_2024.csv` — extracted precipitation features

Verification tip: Plot a quick histogram/map to ensure ranges are realistic (e.g., `prec_annual_sum` not negative; `tmin/tmax` plausible for North Africa).

5) Notebook 3 — `SoilNewDataset.ipynb`

Purpose

- Clip the **HWSD2 raster** to the AOI.
- Join **MU_GLOBAL** pixel codes to **soil attributes** stored in `HWSD2.mdb`.
- Extract **topsoil (D1: 0–20 cm)** properties for modeling.

Expected Inputs

- `ExtractedDatasets/GeoBoundaries/AOI_DZA_TUN/aoi_dza_tun.gpkg`
- `Datasets/SoilDataset/HWSD2_RASTER/HWSD2.bil/.hdr/.prj`
- `Datasets/SoilDataset/HWSD2_DB/HWSD2.mdb`

Key Parameters (typical)

- Layer depth: **D1 (0–20 cm)**.
- Dominant component per mapping unit (highest percentage) when multiple components exist.

Outputs (written to `ExtractedDatasets/SoilFeatures/`)

- `hwsd2_grid01_D1.csv` — topsoil (D1) tabular features per grid cell
- `hwsd2_grid01_D1.gpkg` — vectorized/grid-aligned features
- `hwsd2_grid01_debug.csv` — optional debug/export log of processed rows

Verification tip: Display a few rasters to check units and value ranges (e.g., clay/silt/sand in %, pH around 4–9, OC mostly low in arid regions).

6) Suggested Order of Execution

1. **Boundaries.ipynb** — create AOI and grid
2. **Climate.ipynb** — extract tmin/tmax/prec for AOI/grid
3. **SoilNewDataset.ipynb** — extract D1 soil properties and (optionally) join to grid

If your modeling uses a grid, ensure **all outputs are aligned** (same CRS, extent, resolution, and transform).

7) Common Troubleshooting

- **CRS mismatch / alignment:** Reproject AOI/grid to match rasters ([EPSG:4326](#) for many global datasets). Use `rioxarray.rio.reproject_match()` to snap rasters to a template.
 - **Access/ODBC errors (HWSD2.mdb):** Install the **Microsoft Access Database Engine** (64-bit) and enable the [Microsoft Access Driver \(*.mdb, *.accdb\)](#) ODBC driver.
 - **NoData handling:** Verify `nodata` in `.hdr` and propagate it to outputs when masking/clipping.
 - **Large rasters:** Use chunked reading or dask-backed `xarray` if memory is limited.
 - **Value scaling:** Some datasets store scaled integers—confirm whether you need to apply a scale factor (check metadata).
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8) Deliverables Summary

After running all three notebooks successfully, you should have:

```
ExtractedDatasets/
├─ GeoBoundaries/
│   └─ AOI_DZA_TUN/
│       ├── aoi_dza_tun.gpkg
│       └─ aoi_dza_tun.geojson
├─ ClimateFeatures/
│   ├── grid_0p1.gpkg
│   ├── tmin_2020_2024.csv
│   ├── tmax_2020_2024.csv
│   └─ prec_2020_2024.csv
└─ SoilFeatures/
    ├── hwsd2_grid01_D1.csv
    ├── hwsd2_grid01_D1.gpkg
    └─ hwsd2_grid01_debug.csv
```